Spectroscopic analysis of KIC 8462852

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Abstract

This work analyses the spectrum of KIC 8462852, making use of ESO-MIDAS software for data reduction. We will identify lines present between near-UV and near-IR, with the aim of estimating the Temperature and Spectral Type of the star, as well as calculating Column Density values for optically thin lines.

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I. Introduction

KIC 8462852, also known as Tabby's star or Boyajian's star, is located in the constellation Cygnus at $\sim$ 391pc[?]. It was observed by NASA's *Kepler* mission, whose aim was to measure the brightness of objects in order to discover Earth-like planets orbiting other stars. Its data processing and identification of transiting planet candidates was done in an automated manner through sophisticated computer algorithms, rejecting any data which does not fit in its premises. In this framework, a team formed by amateur and professional astronomers, Planet Hunters project[2], work voluntarily in the task of identifying signals of transiting planets, and have by now discovered almost a hundred exoplanet candidates, including several confirmed systems.

Because Planet Hunter volunteers look at every light curve by eye, serendipitous discoveries are inevitable, and that is how this star appeared on the scene. Studying its flux with time, it was observed to undergo irregularly shaped and aperiodic dips of up to $\sim$ 20% lasting for between 5 to 80 days, which have not an evident answer. Therefore, the astrophysicist community started to think about possible scenarios for this strange behavior that are currently being developed.

Apart from its unusual dips, the star presents irregular, monotonic fading at an average rate of $\sim$ 0.7 mag per century. Photometric studies from the *Kepler* full-frame images found that its brightness monotonically decreased over the four years it was observed, a property not exhibited by none of the 193 nearby comparison stars and the 355 stars with similar stellar parameters used for the study[3].

Among the considered hypothesis are some type of variability, occultation by inhomogeneous circumstellar dust clumps created by catastrophic collisions in an unseen asteroid belt or by a single giant impact in a planetary system, a family of objects on a comet-like trajectory and even a extraterrestrial structure around the star[4;5]. An extensive set of scenarios are being studied, most of which are unsuccessful in explaining the observations in their entirety or ruled out by the lack of IR excess, letting the break-up of one or more massive exocomets as the most compelling explanation consistent with the data at hand by now[2].

Despite its odd behavior, KIC8462852 seems to be a normal main-secuence F3-type star, with a $T_{eff} \approx 6750K$, stellar mass $M_* = 1.43M_\odot$[§], luminosity $L_* = 4.68L_\odot$, and radius $R_* = 1.58R_\odot$[2]

This work analyses the spectrum of KIC8462852 with a first detailed exposition of data reduction with MIDAS in section III. Subsequently, we analyse the lines in the different parts of the spectra between near-UV and near-IR (section IV.1), to finish with an estimation of its temperature and spectral type as well as a calculation of column density values (section IV.2).

[§] $X_\odot$ properties are referred to solar numbers.

II. Observations

Our measurements were taken on March 17 and 18 with the 2.2m Telescope from CAHA Observatory (Almería, Spain), using the CAFOS (=**C**alar **A**lto **F**aint **O**bject **S**pectrograph) instrument. This spectrograph is a focal reducer which changes the 2.2m telescope's f-ratio from f/8 to f/4.4, allowing a shorter exposition time for fainter objects[6].

For spectroscopy, it has a set of different grisms* among which B-200 and G-200 were used[6]. There exist three arc lamps available[7] : HgCdAr, He and Rb, which were used to calibrate our measurements (see figures 3a, 5a).

The Science Spectrum with the grism B200 has an exposure time of 1200 sec, and was taken using USER DEF-1, which reduces the available CCD window. Regarding the one taken with grism G200, it has the same exposure time as before and was taken using USER DEF-2, which results in a longer spectrum as it uses the whole CCD window available. An important detail when collecting data with this last grism is the use of the GG495 filter to avoid line overlapping in the spectra.

Regarding the spectra resolution, each pixel corresponds to $\sim$ 4.5Å and the FWHM calculated from the calibration lines is $\sim$ 2pixels. This means that we can not distinguish between lines closer than 8Å in the spectrum, which makes impossible, for example, to resolve the Na doublet (see section IV. 1c and appendix A).

The data were processed using ESO-MIDAS[8], a system that provides general tools for image and data reduction.

III. Data Processing

To reduce optical spectra several types of images are needed :

- **Bias :** These images are dark pictures which collect all the time-independent electronic noise from the CCD (the reader noise). Although the exposition time for these images is 0 seconds, it is important to assure that its value is not influenced by it. They must be subtracted from the Science Spectra and Flats to correct them from possible systematical errors from the CCD.

- **Flats :** They measure the performance of each pixel in the CCD at different wavelengths, and are used to correct the image from optical defects. They must be taken using the same configuration as the Science Spectra, and with a continuum light (the dawn or dusk light is customarily used as well as a lamp shining at a white wall).

- **Calibration :** These images are produced with the arc lamps and each grism by the dispersion of light, giving not only the variation of the wavelengths in the CCD but the wavelength resolution of the spectra. Like the Flats, they must be taken

*Diffracting devices made up of a prism and a grating.

using the same configuration as the Science Spectra, and the same slit width in particular.

- **Science Spectra :** They are our measurement of the object to study. These spectra need to be processed and corrected using the previous data.

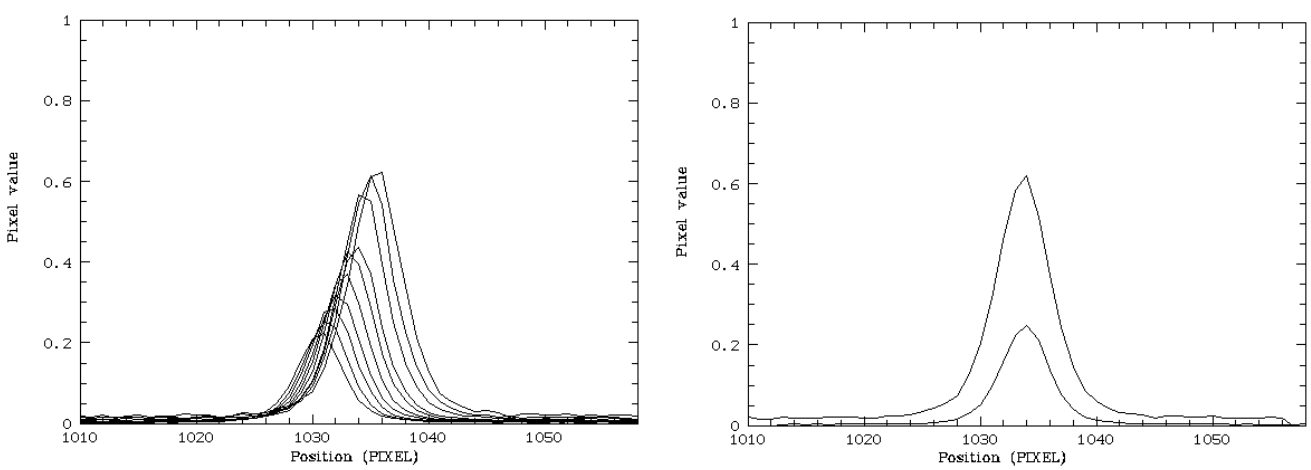
1. Optical spectra reduction

After noise subtraction, applying Bias and Flats' corrections†, the next step consists in extracting the spectrum region from the Science Spectra and Calibration. Since the direction of dispersion is not perfectly perpendicular to the CCD, the spectra appear slanted. For this reason it is important to extract bigger regions than the one occupied by the spectra and proceed to rotate them afterwards. The Calibration image has the same problem, so it needs to be rotated the same angle θ.

To calculate the rotation angle needed, we plot together different rows from the spectrum and we count how many columns the intensity peak has moved in the pixel map as the row is changed (see figures 1a and 2a), following the equation:

$$\theta = \arctan\left(\frac{\text{displacement in pixels}}{\text{distance between rows}}\right) \qquad (1)$$

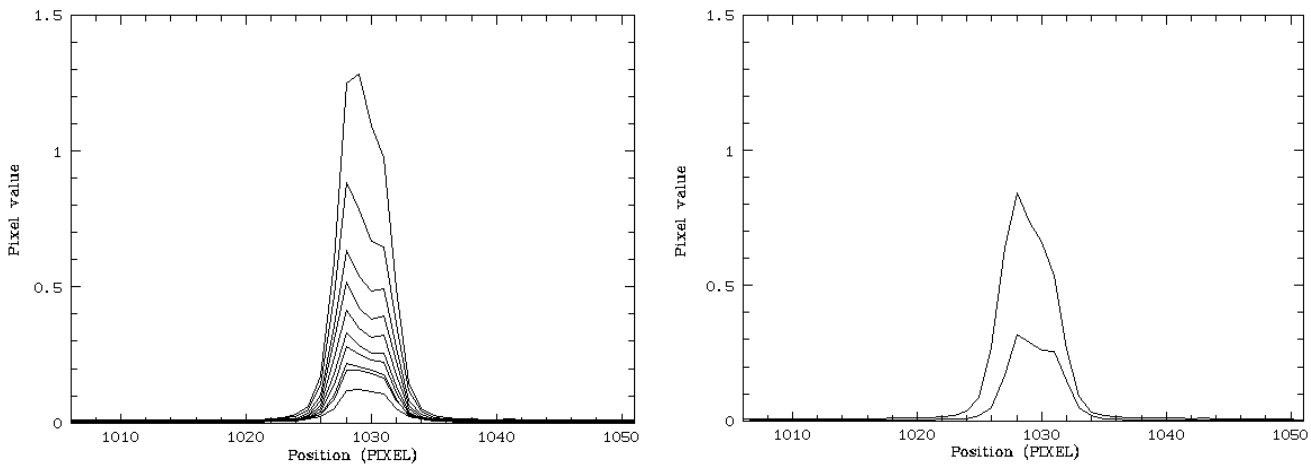
In the case of grism B200, the angle corresponds to $\theta = 0.636593°$ (figures 1a and 1b), while for grism G200 it is $\theta = 0.0716197°$. This last tiny angle appears almost unnoticeable, and there is thus not much difference between figures 2a and 2b.



(a) **(b)**

Figure 1: Grism B200 **a)** Different rows from the spectrum are plotted. A displacement of 5 pixels each 450 rows gives an angle of $\theta = 0.636593°$. **b)** Shows the spectrum rotated. Now the intensity peaks are located in the same pixel for each row.

†As the CCD was not changed during data collection, the Bias used is the same for both spectra. On the other hand, the Flats are specific for each grism to account for their optical properties.



(a) **(b)**

Figure 2: Grism G200 **a)** Different rows from the spectrum are plotted. A displacement of 1 pixels each 800 rows gives an angle of $\theta = 0.0716197°$. **b)** Shows the spectrum rotated.

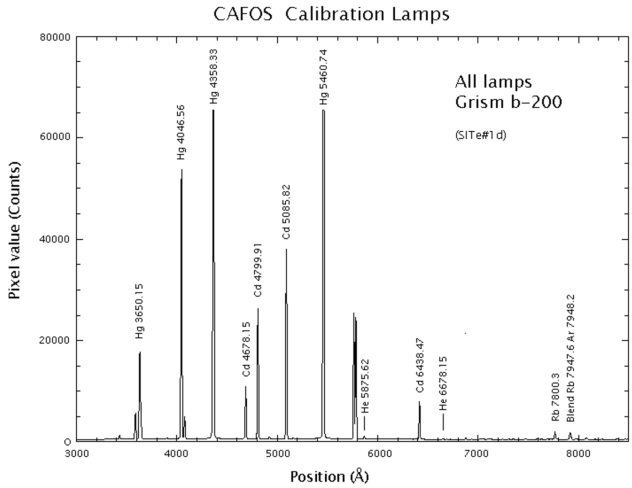
After rotating the Science Spectra and Calibration the problem with the region to extract disappears, so we can proceed with it. We take between $5 - 7$ pixels (approximately the spectrum width) centered on the intensity maximum, and we collapse them into a column to have a one dimensional spectrum, doing the same with the Calibration image.

By now, before starting the spectral reduction part, there is just one thing left to do. The files are arranged so that the wavelength increases from the right to the left, and we need them in reverse order. After inverting the Science Spectra and Calibration we can move to the next step.

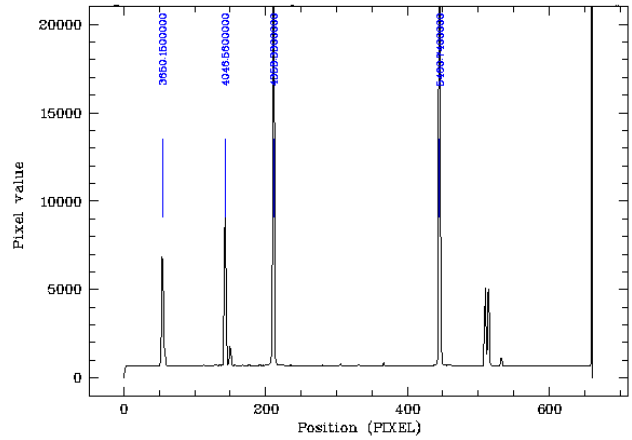
In order to be able to convert pixels into wavelengths, we compare the figures 3a and 3b, and 5a and 5b, and identify the lines that appear in Calibration, as well as their correspondent pixel position. That make it possible to convert both the Calibration and Science Spectra (see figures 4a and 4b, and 6a and 6b).

To ensure everything was made correctly, we compare the accuracy between the position of the lines in figures 3a and 4a, and 5a and 6a. Our data are less intense in both cases, this may be due to a too short exposition time (1sec) or because we did not wait enough after turning on the lamps to take the measurements (It is recommended to wait at least 1 minute before taking arcs) [7]. There must also be a problem with Cd lamp, as none of its lines were seen in measurements taken with grism B200, which forced us to use only the Hg lines for calibration.

With regard to calibration lamps lines for grism G200, at first time, lines corresponding to Ar were omitted in the identification, as in the file descriptor was written so that only HgCd, He and Rb lamps were turned on. Nevertheless, as we said in section II, the HgCd lamp has Ar too, hence in figure 5b these lines appear. However, they were not considered for the spectral reduction.

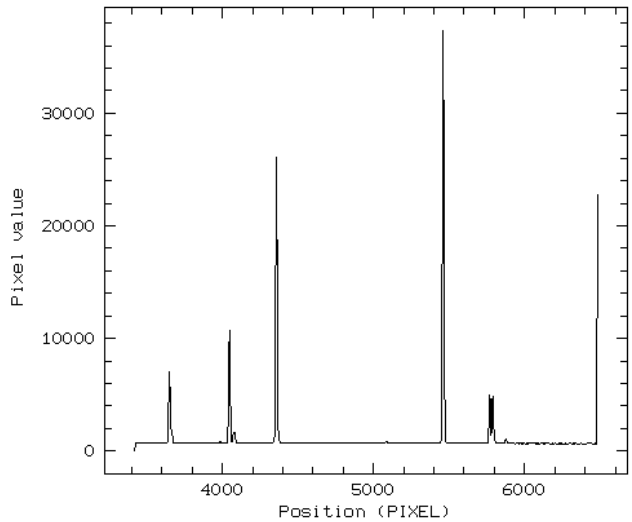


(a)

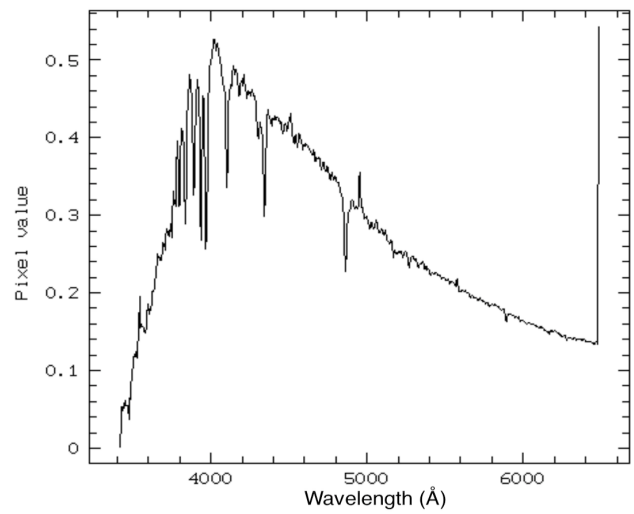


(b)

Figure 3: a) Shows the lines from CAFOS calibration lamps for grism B200[7]. **b)** These are our manually identified lines comparing to the ones from CAFOS.

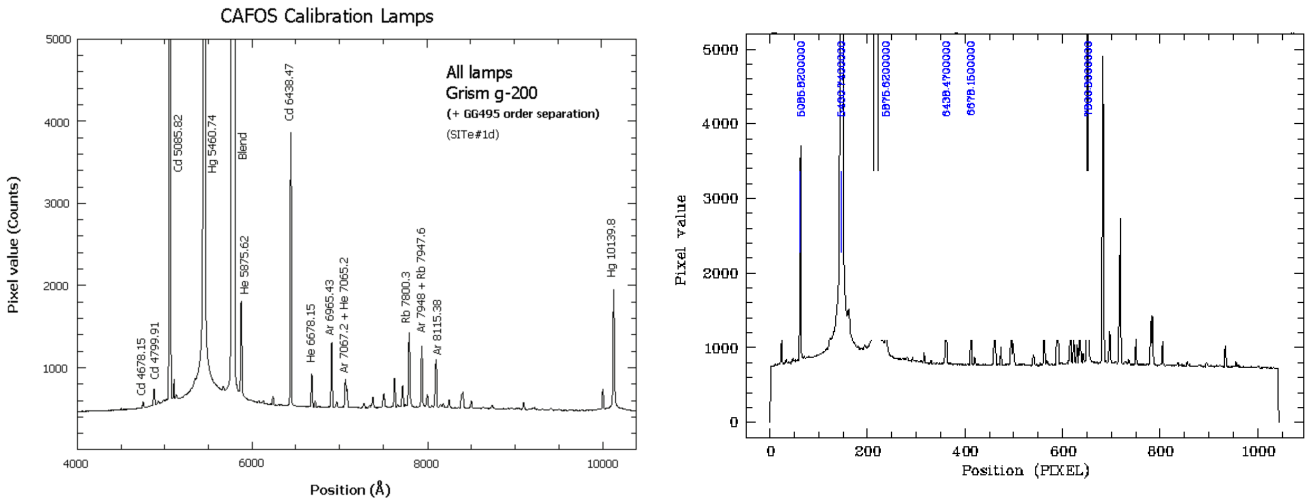


(a)



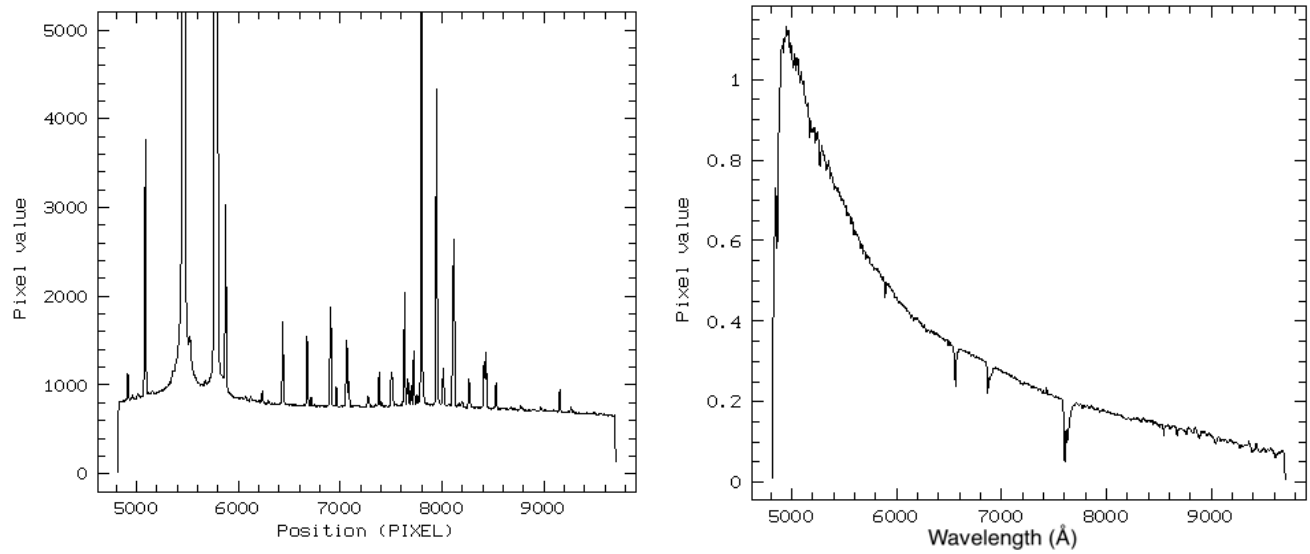
(b)

Figure 4: It shows the pixel value (number of counts in the ccd) for each wavelength. **a)** Corresponds to the Calibration image. We can see the correspondence between its lines and that ones shown in figure 3a. **b)** It is the spectrum of the star (KIC 8462852) without normalization.



(a) **(b)**

Figure 5: a) Shows the lines from CAFOS calibration lamps for grism G200[7]. **b)** These are our manually identified lines comparing to the ones from CAFOS.



(a) **(b)**

Figure 6: It shows the pixel value (number of counts in the ccd) for each wavelength. **a)** Shows the Calibration image. We can see the correspondence between its lines and those ones shown in figure 5a. **b)** It is the spectrum of the star (KIC 8462852) without normalization.

IV. Experimental Results

1. Spectra analysis

Before proceeding with the analysis of the absorption lines consists in normalizing the spectra by dividing it into sections, in order to improve the accuracy. This step is made by hand, selecting the continuum from a plot of the spectrum.

Due to earth relative motion toward the star, the spectra appears moved. It can be corrected by calculating the barycentric velocity, which gives us the wavelength's displacement $\Delta\lambda = \lambda_0 \frac{v}{c}$, where λ_0 corresponds to the line's real wavelength. To compute this velocity, the date and Universal Time were taken from the image's header, together with the RA and DEC of the star $(20h06m15.457s\ +44°27'24.6'')$[9] and observatory's latitude and longitude $(37°13'15.0"N\ 2°32'48.4"W)$[10]. The obtained results were:

$$v_{B200} = 7,949709 km/s$$

$$v_{G200} = 8,156923 km/s$$

The displacement they produce varies between $(0.1 - 0.24)$Å, so that it was not taken into account as it is far below the spectral resolution (see section II).

On the other hand, to reduce the human error selecting the intensity peak for each line, this was computed by gaussian profiles.

We will focus on analysing the spectrum of KIC 8462852, that as we said before, is a star of class F3V. We will try to prove that by comparing the typical characteristics of these stars with ours.

a Balmer series

Balmer series refers to Hydrogen transitions from $n \geq 3$ to $n = 2$ (where n designates the principal quantum number of the electron). In figure 7 we can see the Balmer lines[13;14;15] for $n = 12-3$ where the Hε $(7 \to 2)$ is blended with ionized calcium (CaII H) as their wavelengths are quite close.

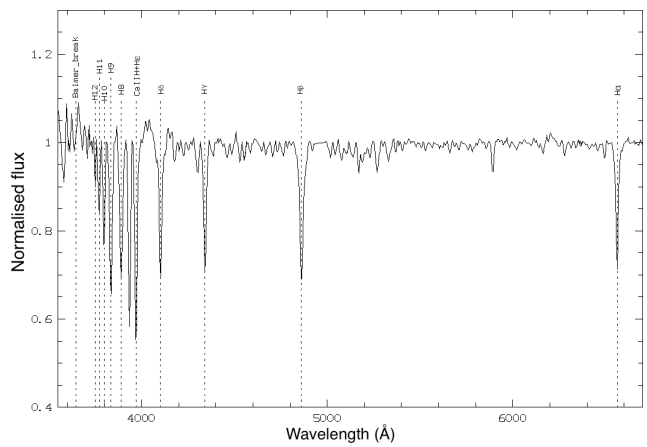


Figure 7: Balmer series containing the Hydrogen transitions from $n \leq 12$. The Hε $(7 \to 2)$ is blended with ionized calcium (CaII H). It shows the Balmer break at a wavelength of $3646\mathring{A}$[11;12].

These transitions stand out among the rest of the spectrum (apart from telluric O_2 bands), having the highest equivalent width values (see appendix A).

b The Ultraviolet

The analysis starts beyond the Balmer break. The first aforementioned Hydrogen transition found corresponds to H12 $(12 \to 2)$ line, some

other Balmer lines and two ionized Calcium lines (CaII K and CaII H+Hε) (see figure 8).

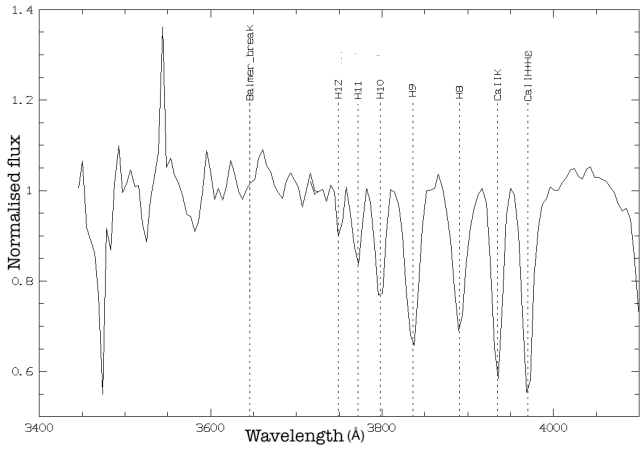


Figure 8: Ultraviolet part of the spectrum. It shows some Balmer lines as well as the Balmer break and two ionized Calcium lines.

c Visible Spectrum

This section holds most of the analysed spectrum of the star, with well known lines such as Hβ and Hα, the sodium doublet (NaI D) and the B-band (O_2) from the Earth's atmosphere.

Together with Hydrogen lines, the metallic-line spectrum of this part is usually used to determine the temperature of the stars. In addition, the G-Band due to the CH diatomic molecule, which can be suspected (figure 9), is also suitable for it as it grows rapidly with declining temperature and is characteristic of stars cooler than F3[13].

Studying the blue part of the spectrum (figure 9), uncertain metal lines‡ have been identified by comparison with a model spectrum for stars of class F2V and F5V[13]. They are in the expected place but the sigma of their line depth is within the statistical error of the continuum, hence they are marked as a possible lines (this method will be used along the analysis to identify most of the lines marked as uncertain lines). The three deepest lines correspond to Hydrogen transition from Balmer series (Hδ, Hγ and Hβ).

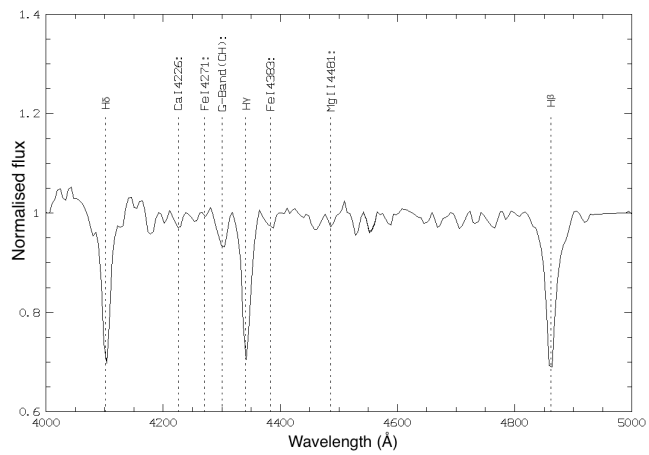
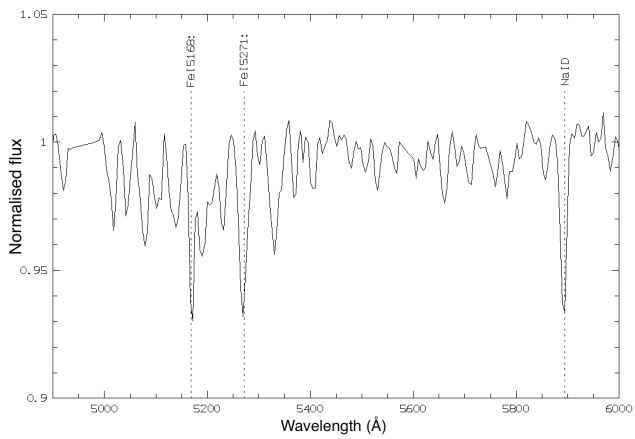


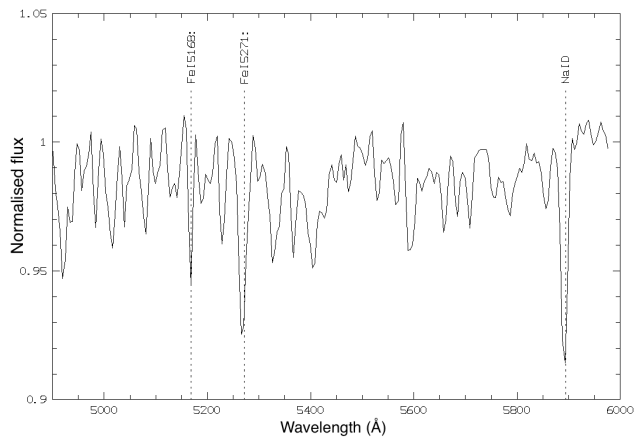
Figure 9: Blue sector shows three Hydrogen transitions (Hδ, Hγ and Hβ) as well as some possible metal lines and the G-Band (CH).

Figures 10a and 10b represent the same part of the spectrum, taken with each grism, allowing us to recognize some metal lines as well as the NaI Doublet from the background noise[14;15]. Their intensity is not so different from the continuum but they are present in both images, making their identification possible.

‡Uncertain lines are indicated as *Line name:*



(a)



(b)

Figure 10: Yellow and green part of the spectrum. It shows two uncertain FeI lines and the NaI Doublet among the background continuum whose identification has been possible by comparison of both images. **a)** Spectrum taken with the grism B200 **b)** Spectrum taken with the grism G200.

Last part of the visible spectrum (figure 11) shows the Hydrogen transition Hα[14;15] and a telluric band originated from absorption by oxygen molecules in the Earth's atmosphere (the Fraunhofer line called O_2 B-band[16]). At around $\lambda = 7200\AA$ a H_2O band[16] seems to appear, but zooming in to compare with noise background, it is not unambiguous enough (figure 12).

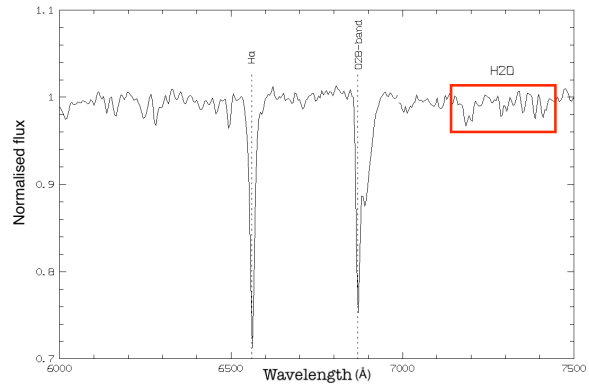


Figure 11: Orange and red sectors show the Hydrogen transition line Hα from Balmer series and a telluric band called B-band (O_2) caused by Earth's atmosphere. An uncertain telluric H_2O band has also been tentatively identified.

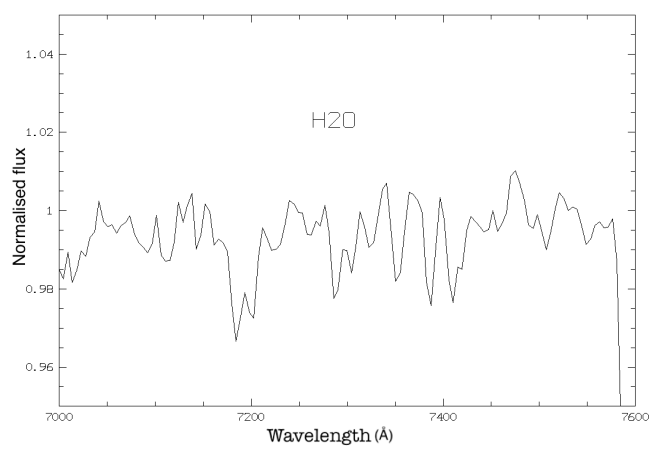


Figure 12: H_2O band. There is not an appreciable difference between noise background and the possible band.

d The Near-infrared

This subdivision contains lines from telluric contamination[16] (figure 13), Earth-origin sources from absorption by the molecules in the atmosphere. By comparison with the previous said model spectrum for stars of class F2V and F5V[13] some uncertain metal lines, the Calcium triplet and transitions belonging to

Paschen series have been identified among the telluric bands (figure 14).

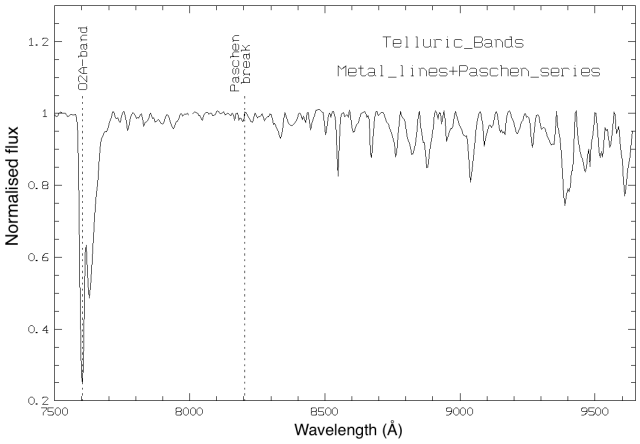


Figure 13: Near-Infrared part of the spectrum. It presents telluric bands from Earth's atmosphere and the Hydrogen transitions belonging to Paschen series.

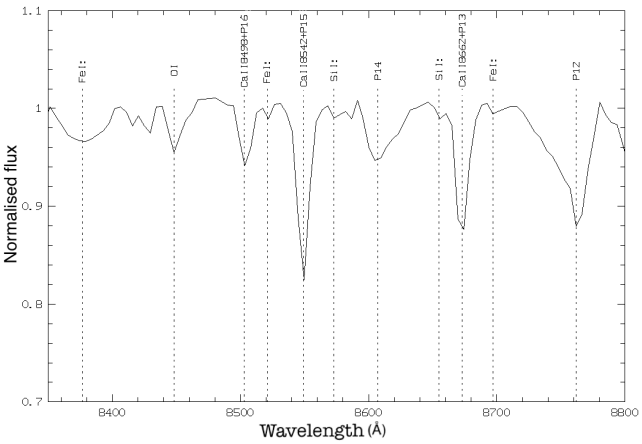


Figure 14: Region of the higher Paschen lines, essentially tellluric-free. It shows the analysis of some uncertain metal lines and the Calcium Triplet, blended with $P13$, $P15$ and $P16$ Hydrogen transitions.

Paschen series consists in Hydrogen transitions from $n \geq 4$ to its third level, whose lines lie all in the infrared part of the spectrum and are labelled as Pn. Some of them are blended with the Calcium Triplet, as it can be seen in figure 14, being a plausible explanation for the reddening displacement of the identified Calcium Triplet. As we mentioned before, the real resolution of the spectra is ~ 8Å, consistent with the $\Delta\lambda$ between Paschen transitions and the ionized Calcium lines, which can not be separated.

2. Analysis of Stellar Characteristics

a Temperature and Spectral Type

Both the temperature of the star and its spectral class can be delimited by analysing its absorption lines, among which we find that our star presents Hydrogen and both neutral and ionized metals (eg. FeI, CaII ... in figures 8, 10 and 14). Studying the strengths of absorption lines curves combining the effect of Saha and Boltzmann [18], the absence of FeII and TiO, and the identification of Hydrogen, CaII and FeI determine a stellar surface temperature around 7000K, which is consistent with a F-type star.

It was mentioned that the identification of a G-band (CH), as we observe in figure 9, means also a F-type star beyond F3.

In addition, there exist some studies relating equivalent width of some lines with spectral type of stars, which provide a more accurate method to determine these characteristics. The work developed by Danks et al. (1994) [19] gives a re-

lation between equivalent widths[§] of Hα, P_{12}, OI (7776Å), CaII (8662Å) and NaI doublet with spectral type and luminosity class.

The analysis of those lines in our spectrum (see EW in Appendix A) match in most of the cases with a F0-F5 type star. Our measurement for Hα is in between a F0-F5 type star, P_{12} is not well correlated, probably due to its strange profile which produces a higher EW value, OI (7776Å) was not observed in the spectrum, EW of CaII (8662Å) corresponds to a star in between of F0-F5 type too, and finally, the measurement of NaI doublet, neither separated in their spectra, match with a $\sim$ F2-type star.

Putting all these information together, it could be determined that KIC 8462850 is a star of F3-F5 type, being presumably in main-sequence luminosity class.

b Column density values

Column density[¶] values can be obtained, assuming an optically thin case (none of the lines reach the zero intensity) by[21;22]:

$$W_\lambda = \frac{e^2\pi}{m_e c^2}\lambda^2 f N_\lambda \tag{2}$$

where $\frac{e^2\pi}{m_e c^2} = 8.853 \cdot 10^{-13}$cm, for W_λ and λ expressed in cm, and N_λ in cm^{-2} (see Appendix A for values).

In order to check if we are in this regime, we need to look at oscillator strength and line-depth ratios. If its quotient remains approximately constant for each line, it means they are optically thin, otherwise it is only possible to determine a lower bound for column density values. Looking at figure 15, the behavior of CaII K, CaI, MgII, Hα and OI do not follow that of the rest, which is a possible indicator of optically thick lines.

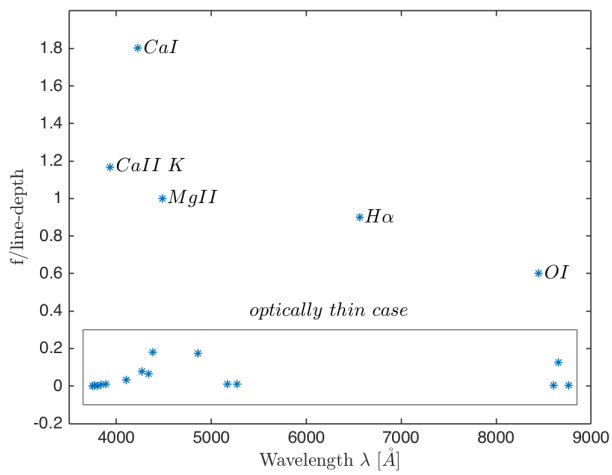


Figure 15: Ratio of oscillator strength to line-depth for each wavelength. Lines inside the box correspond to optically thin cases whereas those outside does not match with this behavior.

In the case of CaI and MgII, they are lines marked as uncertain, so an immediate explanation could be a misidentification. On the other hand, CaII K is the deepest not blended line, making it plausible to be optically thick. The cases of Hα and OI, are possibly due to

[§]In this paper, EW are said to be measured in mÅ, but the good correspondence with our measurements and the fact that their spectral resolution is said to be 4.5Å per pixel (similar to ours) make that unit appear to be a mistake. Moreover, there exist other studies[20] where EW is measured in Å with a good correspondence with ours. Nevertheless, unfortunately, it was impossible to access to the related bibliography mentioned in the paper to prove it.

[¶]Column density refers to the number of atoms along the line-of-sight.

its high oscillator strength value (also another explanation for CaI and MgII), which makes the quotient to be higher than the rest.

V. Conclusions

This paper shows a spectroscopic analysis of KIC 8462851, a F3V star with a temperature of ∼6750K.

The wavelength range available has allowed us to study both the Balmer and Paschen series (the last one only partially), as well as lines such as the NaI doublet in the visible spectrum or the Calcium triplet in the near-IR.

Due to the low spectral resolution (8Å) and the lack of more lines was impossible to determine a radial velocity for the star. However, the presence of Hydrogen and both neutral and ionized metals has been used to estimate a temperature of around 7000K, idea reinforced by the study of their EW, which makes it match with a F0-F5 type star. This range is also consistent with the presence of the G-Band (CH) identified.

Regarding the study of column density values for each line, it was necessary to ensure an optically thin case for each of them to use equation 2, which was presumably achieved for the vast majority (with only 5 exceptions).

A. Lines' characteristics

Line	λ_0[Å][15;17]	λ_{obs}[Å]	EW[Å]	f[15;17]	$N_\lambda[cm^-2]$
H12	3750	3749	0.860851	0,00021521	$3,21301 \cdot 10^{16}$
H11	3771	3772	1.98610	0,0028368	$5,5612 \cdot 10^{15}$
H10	3798	3798	3.31923	0,00038526	$6,74657 \cdot 10^{16}$
H9	3835	3836	6.17225	0,0054317	$8,72743 \cdot 10^{15}$
H8	3889	3890	5.89655	0,0080397	$5,47762 \cdot 10^{15}$
CaII K	3935	3935	5.89443	0,682	$6,30489 \cdot 10^{13}$
CaII H + Hϵ	3970 + 3970	3970	7.45976		
Hδ	4102	4102	7.83159	0,022105	$2,37836 \cdot 10^{15}$
CaI	4226	4226	0.422293	1,75	$1,52625 \cdot 10^{12}$
FeI	4271	4271	0.0487893	0,0762	$3,96479 \cdot 10^{12}$
G-Band(CH)		4301	1.34859		
Hγ	4341	4341	5.43493	0,044694	$7,28911 \cdot 10^{14}$
FeI	4383	4383	0.550104	0,176	$1,8378 \cdot 10^{13}$
MgII	4481	4486	0.265179	0,981	$1,52065 \cdot 10^{12}$
Hβ	4861	4862	8.17687	0,11938	$3,27426 \cdot 10^{14}$
FeI	5168	5168	0.548611[‖]	0,00847	$2,73934 \cdot 10^{14}$
FeI	5271	5271	1.36477[‖]	0,00916	$6,05742 \cdot 10^{14}$
NaI D	5891 + 5897	5894	1.153875[‖]		
Hα	6563	6561	5.18575	0,64108	$2,12131 \cdot 10^{13}$
B-Band (O_2)	6870	6870			
A-Band (O_2)	7600	7604			
~~FeI~~		~~8377~~	~~1.16283~~		
OI	8448	8448	0.635108	0,574	$1,75121 \cdot 10^{12}$
CaII + P16	8498 + 8502	8503	0.621635		
~~FeI~~		~~8521~~	~~0.0652772~~		
CaII + P15	8542 + 8545	8549	1.98357		
~~SiI~~		~~8573~~	~~0.0658977~~		
P14	8598	8607	1.52865	0,0022246	$1,04996 \cdot 10^{15}$

¶‖B200 and G200 average value.

Line	λ_0[Å] [15;17]	λ_{obs}[Å]	EW[Å]	f [15;17]	$N_\lambda[cm^-2]$
SiI	8651	8655	0.0494699	0,125	$5,97321 \cdot 10^{11}$
CaII + P13	8662 + 8665	8673	1.62594		
~~FeI~~		~~8697~~	~~0.0780531~~		
P12	8750	8762	3.42014	0,0037133	$1,35887 \cdot 10^{15}$

Table 1: It shows some details about the identified lines in the spectrum. Line refers to line name (chemical element and ionization yield), λ_0 is the line wavelength in vacuum while λ_{obs} is the observed wavelength from the spectrum, EW correspond to equivalent width, f is the oscillator strength and N_λ, the column density. The lines that are crossed correspond to those firstly classified as uncertain but eliminated from the analysis for different reasons afterwards (poor coincidence with λ_0, small value of EW...). Telluric bands are not relevant for the study of this star spectrum and they only show information about λ_0 and λ_{obs}.

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